

The AUDI AG Quality Manual



LAH 893 010

Version **2.5**
29. August 2006

Nina Kandler-Schmitt
AUDI AG Quality Assurance
Phone: +49 (0) 841-89-40765
E-Mail:
nina.kandler-schmitt@audi.de

A handwritten signature in black ink, reading "Zimmermann", is enclosed in a rectangular box.

W. Zimmermann
Head of Quality Assurance
AUDI AG

Contents

Introduction.....	3
Terminology/abbreviations.....	4
1. General requirements	7
1.1. Collaboration of supplier.....	8
1.2. Nominating a person responsible for quality assurance.....	9
1.3. Overall responsibility for quality / compliance with quality assurance requirements at sub-suppliers	9
2. Quality planning.....	10
2.1. Quality planning design / pre-production.....	10
2.2. Quality planning series production.....	11
3. Parts lifecycle, identification of parts, quality certificate for pre-production.....	11
3.1. Parts lifecycle.....	12
3.2. Identification in the pre-series phase.....	12
3.3. Quality certificate for the pre-production phase.....	13
3.4. Marking at changes in the series.....	14
4. Master jig harmonisation.....	14
4.1. Requirements for master jig harmonising.....	14
4.2. Conducting measurements and measurement procedure	15
4.3. Data flowback.....	15
4.4. Tooling/part design.....	15
5. Process capability prior to and during series production.....	15
6. Internal process and product audits	16
7. First sampling	16
7.1. First sampling and approval requirements.....	16
7.2. Contents of first sampling documents.....	17
7.3. Addresses where first samples are to be sent	19
8. Pre-production (2-days' production).....	20
9. Quality backup during series production.....	20
9.1. 0-km complaints.....	20
9.2. Reworking and sorting action.....	21
9.3. Analysis of field complaints.....	21
9.4. Continuous quality monitoring.....	22
9.5. Audits, Problem analyses and Technical Revision Supplier (TRL)	22
9.6. Media	22
9.7. External service providers.....	23
10. Attachments	24

Introduction

The AUDI AG Quality Manual describes the general requirements for suppliers of purchased parts from the point of view of Quality Assurance. Requirements relating specifically to parts can be found in the Technical Development Components Manual or in the drawing.

The supplier, upon submitting the offer, confirms that he is familiar with the requirements described in the Quality Manual, accepts these and will apply them and comply with them without reservation.

The German-language expenditure of the Audi Quality Manual is obligatory.

Edition 2.5 which has been sent to you, has been updated compared to Version 2.4.

[Changes/Innovation of the current version 2.5 in relation to version 2.4](#)

Chapter	Change/Innovation	Page
Terminology/Abbreviations	Supplementation of the Term "set parts"	4
Reference sources	Software qs-Stat 3.x and Guideline	5
1	Textual supplementation	
1.3	Textual supplementation	
2.1	Textual supplementation	
3.2	Textual supplementation	
3.3	Textual supplementation	
3.4	Textual supplementation	
5.	Textual supplementation	
7.1	Textual supplementation	
7.2	Declaration of forerunner sampling by remark on cover of sampling documentation	
7.3	Textual supplementation	
9.1	Textual supplementation	
9.3	Textual supplementation	
9.4	Textual supplementation	
9.5	Textual supplementation: Technical Revision Suppliers	
Attachment 8	New Version of "Änderungsdreieck"	
Attachment 10	New Version of the Checklist	

Further new sections/revisions compared to Version 2.4 are recognisable from the blue lettering.

Terminology/abbreviations

0-Serie	Direct forerunner of series production to optimise dimensional accuracy and function of the production components.
2TP	2 days' production (pre-production)
AE	Permission of deviation
Supplied products/parts	Components/products provided to the supplier by AUDI/ Volkswagen for further processing/finishing at an agreed price or free of charge
BMF	Design approval
BTV/TV	Person in Charge of AUDI AG for certain component/product
CAD	Computer Aided Design
C _{mk}	Machine Capability. Index taking into account the position
COP	Conformity of Production
C _{pk}	Process Capability. Index taking into account the position
D/TLD Parts	Parts of which one or several characteristics are subject to mandatory documentation. Compare also VDA Volume 1 term "DmbA" (documents requiring special filing)
EMPB	First sample test report
Gen.	Generation status
HW/SW	Hardware / software
i.O.	OK
KWZ	small series tool
P _{pk}	Preliminary Capability. Taking into account the position
ppm	Parts per million. States the ratio of defective parts related to 1 million supplied parts within a defined period.
PVS	Series production testing
PV-Serie	Series production testing
Q	Quality
QFD	Quality Function Deployment
QM	Quality management
QPN	Qualification programme for New Parts / Formula Q, New Parts - integral
QS	Quality assurance
QTS	Quality data information and processing system
RPS	Reference point system
SE	Simultaneous Engineering (group of representatives of various corporate divisions involved in the product creation process)
Set parts	Parts which are defined (set) in a framework agreement with third suppliers by AUDI/Volkswagen as single parts or assemblies with annual purchase quantities as well as prices, but which are purchased or scheduled by system/module suppliers. Sampling and release for set parts has to be done by the responsible supplier of assemblies.
SOP	Start of Production
SWZ	series tool
TL	Technical specifications for delivery
TLD	Technical directive documentation
VDA	German Automotive Industry Association
VFF	Pilot lot release vehicles: with series tool falling parts 8 weeks before PVS
ZSB	Assemblies

For other terminology specifications, refer to Formel Q-konkret.

Only for information! The German-language expenditure of the AUDI Quality Manual is obligatory.

Associated documents

For all applicable documents and forms the newest valid expenditures are obligatory.

- VDA volumes
- Formel Q-konkret
- Formel Q-Capability
- Q-Certification in Preproduction
- Formel Q New Parts – integral (Qualification Programme for New Parts (QPN))
- VW 01055 Reference Point system, RPS, Drawings
- VW 01058 Technical Drawings, Lettering
- VW 01154 Development Conditions, General Requirements
- VW 01155 Vehicle Supply Parts, General, Approval of First Supply and Changes
- VW 10119 Prüfprozesseignung, Berücksichtigung der Messunsicherheit in Prüfprozessen
- VW 10130 Maschinenfähigkeitsuntersuchung für messbare Merkmale
- VW 10131 Prozessfähigkeit für messbare Merkmale
- Purchase conditions for production material
- - 8D Report conforming to VDA/QMC (*Internet-Link: <http://www.vda-qmc.de>*)
- - LAH 893 012 Lastenheft Umformtechnik der Audi AG
AUDI AG manual for forming technology
(Applies to formed parts, weld groups and sheet metal assemblies)
- - LAH 893 018 Lastenheft Oberflächenanforderungen an lackierte Außenteile
Surface requirements in respect of painted exterior parts Manual
(Applies to exterior parts which are supplied painted)
- - LAH 893 019 LAWA Manual
(Applies to parts which are painted in the AUDI paintshop for colour-keyed attached parts)
- - LAH 893.909.A Requirements for Software
(Applies to Software and Software of certain controll units in the car)
- - LAH 893 880 Sicherheitsausstattung Airbag
- - LAH 893 857 Sicherheitsgurte

Reference sources

VDA volumes: <http://www.vda-qmc.de>
Formel - Q: <http://www.vwgroupsupply.com>
Standards: <http://www.vwgroupsupply.com>
Manuals: <http://www.vwgroupsupply.com>

Software qs-STAT 3.x and Guideline "Fähigkeitsnachweis von Messsystemen":

Q-DAS GmbH : Eisleber Str. 2, D-69469 Weinheim, Tel. 0049 (0)6201 3941-0,
e-mail: q-das@q-das.de

Note:

Current information can be found on the Internet (<http://www.vwgroupsupply.com>) for the entire VW Group under Group Quality Documents and Audi specific information under Brand Quality Assurance-Audi.

Updates of Audi AG's Quality Manual will be announced via the Internet
(<http://www.vwgroupsupply.com>) under News.

1. General requirements

AUDI AG Quality Assurance requires components to be supplied on time and according to the specifications of the drawing. During the lifecycle of the components, they have to satisfy all functional, constructional, dimensional, physical, chemical, kinematic as well as environmentally compatible and visual requirements, in accordance with the standards. The following requirements must be complied without any reservation.

- Certification conforming to VDA 6.1 or ISO TS 16949:2002.
- Auditing of the production site by one of the VW Group auditing departments in accordance with the Formel Q Capability with at least a 92 % degree of compliance, "A" rating at the time of order release. In exceptional cases, a supplier with a "B" rating (at least 82 % degree of compliance) can be considered, if the specified improvement measures have been implemented by the supplier and an "A" rating is achieved within 12 months but no later than the SOP (start of production).
- A Quality Management as described in Formel-Q is implemented and used.
- Group D parts release in conformity with the Formel Q-Capability list of requirements for D/TLD parts.
- Compliance with the ppm objectives "Zero fault strategy". The current ppm values for 0km can be obtained from the QTS system. The ppm values for "Feld" can be taken from the "Kaufteile-Schreibungsliste" which is forwarded to suppliers monthly by I/GQ-32 "Regressierung". As if the ppm objectives are not reached for existing series production supplies, the AUDI AG taskforce together with the supplier will draw up a plan of action for achieving the group target.
Before new contracts are placed, ppm-targets are agreed upon with the supplier. Deviation from the ppm-targets stated with the bid is only allowed when agreed upon a separately Quality-Agreement with the Audi Quality Assurance/ "Produkttechnik". Differing targets in bids will be meaningless.
With start of series delivery a ppm-programm for 0km and "Feldbeanstandungen" will be stipulated and started to ensure continuous quality improvement in series.
- Compliance with legal environmental regulations in force.
- Compliance with legal regulations in force
Supplier is liable to ensure the compliance with the legal regulations in force concerning the individuell market and has to give prove about this to AUDI AG if demanded. Deviations (e.g. loss of accreditation or certification) has to be reported to AUDI AG.
Contacts for „Conformity of Production“ (COP) in the plants are:
 - Ingolstadt:
Mr. Handl Tel. 0049 (0)841-89-89652 ludwig.handl@audi.de

- Neckarsulm:
Mr. Krieg Tel. 0049 (0)7132- 31-5074 manfred.krieg@audi.de
- Győr:
Mr. Kovacs Tel. 0036-96-66-1139 istavan.kovacs@audi.hu

➤ Link (online connection) to the QTS quality data information and processing system.

- *QTS Hotline: Tel.: +49 (0)180 5 729 433 (0,12€/min), e-mail: qts-online@audi.de*

- *QTS Training: Tel.: +49 (0) 841 96 602 520; e-Mail: akademie.mueller-weitkunat@audi.de*

➤ Statement of material composition.

The documentation of the precise chemical composition of the materials used with all ingredients is mandatory. Such data have to be safeguarded by means of the International Material Data System IMDS. By using the on-line sampling system (BeON) the data sheet must be registered in the IMDS system at latest one day before Over night an automatic alignment with the AUDI defaults takes place here. Only with accepted entries the on-line sampling can be passed on to Audi.

- Internet address: <http://www.mdsystem.com>

Requirements for the input of the data are described in the VOLKSWAGEN 01155 or within the IMDS guidelines. The file input for complete parts and ZSBs has to take place similar to the Attachment 9.

➤ Relocations of production or major changes¹ to the manufacturing process must be communicated to AUDI AG as soon as possible and require the approval of AUDI AG Quality Assurance.

A project plan and safeguard concept has to be prepared and agreed with AUDI AG for relocations (including those within a production site). If a relocation is implemented, the parts have to be submitted for sampling to AUDI AG, prototype approval must be obtained again for obligatory prototype parts. A new 2TP has to be conducted after each relocation.

In the event that the supplier relocates components to another production location without being required to do so on the basis of technical constraints resulting from new requirements defined by Audi or without co-ordination and approval of AUDI AG Purchasing Department, the costs that occur at AUDI AG Quality Assurance can be charged to the supplier.

In the event that component relocations are not advised or are not advised timely to AUDI AG in written form, supplier may be barred for new orders.

If the requirements of the AUDI AG Quality Manual are not completely fulfilled the costs incurred in such a case by Quality Assurance may be charged to the supplier.

1.1. Collaboration of supplier

¹ Changes of the delivering-subject, for example the constructive carrying out, the material and his composition, the manufacturing process, the sub-suppliers and/or the production plant

Active, independent and innovative collaboration is expected from the supplier. The supplier has to take it upon himself to improve his processes and to define and introduce measures that are effective in the short-term and long-term and result in capable processes. The specified measures have to be implemented in accordance with the schedule.

The supplier must ensure that a contact person competent in German is available and that correspondence (e.g. EMPB, 8-D-Report) is carried out in German.

1.2. Nominating a person responsible for quality assurance

At the start of the project, the supplier has to nominate a competent person who has the authority to **represent the quality assurance** interests of the supplier in the SE and project work.

The name, function and representative has to be advised at the start of the project to the SE technical group and to the [BTV/TV](#).

1.3. Overall responsibility for quality / compliance with quality assurance requirements at sub-suppliers

Acceptance of total responsibility for component quality by those suppliers who have direct contractual relationships with AUDI AG. Where there is more than one contractual partner, the overall responsibility is assumed by the one delivering the final assembly to the Audi plant.

The system supplier (assembly supplier with assembly operations and responsibility for individual components/assemblies) is also responsible for sub-suppliers compliance to quality assurance requirements.

This includes the following points (amongst others):

- Assuring the quality capability of sub-suppliers²
- Verification of quality performance and quality capability of sub-suppliers
- Upgrading the qualification of sub-suppliers
- Defining quality assurance agreements (also applies for set parts). Appropriate attention has to be paid to the AUDI AG Quality Assurance requirements
- Assuring the flow of information / forwarding of information between contractual partners
- Specifications relating to handling of D/TLD parts as well as the necessary documentation
- Specifications relating to warranty and traceability of components
- Sampling of all components/assemblies from sub-suppliers
- Safeguarding the realisation of pre-production (2TP) at sub-suppliers before the start of series production (successful application of the “Qualification Programme for New Parts (QPN) (= Formel Q New Parts – integral)”) (*Attachment 1*)
- Assuring the qualitative and quantitative component supply over the product lifecycle
- Application of Formel-Q in respect of sub-suppliers
- High-strength bolts must be procured from suppliers who have been approved by the Volkswagen Group for this purpose (<http://www.vwgroupsupply.com>)

² see also Requirements VW Formel-Q-Konkret Chapter 4.15

Suppliers of modules (set and/or supplied products/parts) are responsible for the quality assurance for all commissioned value creation aspects of the whole module. The module supplier is responsible for quality assurance of supplied and set products regarding assembly and supplied condition.

Set parts are parts and extents, for which the source of supply is given by the Audi Procurement Department. First part approval process [has to be done by the system supplier/ module supplier](#). Deviations and exceptions herefrom have to be [agreed upon supplier and Audi quality assurance and procurement](#). Afterwards deviations and exceptions are dealt separately in the contract.

2. Quality planning

2.1. Quality planning design / pre-production

The supplier has to prepare a time schedule which contains all product and quality related deadlines as well as the project deadlines specified by the customer (e.g. SOP, 0-series, PVS, quality safeguarding procedure, FMEA, inspection times, BMF, sampling, 2TP, approvals).

The supplier is responsible for conducting suitable QFD analyses, feasibility studies, “zero fault strategies“, design and process FMEA’s (on request also in German/English) to create and implement quality assurance and measurement concepts (such as statistical tolerance chain analysis including the effects of attached parts and assembly variability), process flowcharts, production resources and maintenance scheduling.

In producing the FMEA’s account has to be taken of experience from identical/comparable products and processes (expertise memory). The design FMEA has to be presented to AUDI AG Technical Development and the process FMEA to AUDI AG Quality Assurance.

As a part of the development activity the supplier has to define important quality characteristics of the product and the process. They have to be agreed with the customer, for example as part of the SE collaboration. Any existing customer standards have to be taken in priority -. The supplier has to create inspection schedules and has to submit these to AUDI AG Quality Assurance. The inspection characteristics of the product and process which have been derived from the defined quality characteristics, have to be documented in these inspection schedules. This applies in particular to D/TLD parts for which all mandatory documentation characteristics must be contained in inspection schedules.

Electrical components, assemblies and functional parts must undergo a 100% inspection in respect of function of the supplied unit. Concerning these scopes of supply the Attachment 10 is to be filled out and to be handed over to the responsible person of the AUDI AG quality assurance within four weeks after disposal of the A-samples -. Up to the delivery of the B-Sample³ the manufacturing concept is to present to the AUDI AG quality assurance department on the basis of production workflow, the control plan and the manufacturing layout-. Before

³ See Definition of sampling phase in component specifications (Development Department), Chapter 2: Definition

the procurement of series production facilities and series testing facilities the manufacturing concept has to be co-ordinated and to be specified with the AUDI AG quality assurance. Separate tests must be carried out by the suppliers for COP relevant content (legislation in force) in conjunction with the COP contacts at the AUDI AG plants. These must be documented accordingly and the OK test results must be forwarded to the COP contacts at AUDI AG.

The supplier has to provide or procure suitable inspection and test resources up to PVS in order to inspect and assure all functional, installation and quality-related characteristics. Inspection and test resources (e.g. gauges, [equipment models](#), [measurement records](#)) relating to a specific project must be included and precisely specified in the quotation. It is assumed that general standard test resources (3D measuring machine, colorimeters etc.) are available. The inspection and testing capability of all inspection and test resources used has to be verified by examinations up to the start of the 0-series. Verification has to comply with the general Guide of the European Automobile Industry in respect of the "Verification of Capability of Measuring Systems". This is available from Q-DAS.

As part of quality responsibility for complete systems, [which are assembled in an pre- or end assembly process to a ZSB \(such as the complete cockpit or panelling parts connected to the sky\)](#), appropriate [additional](#) assy inspection devices (e.g. [parts of](#) master jigs, [mounting and demounting cubes](#)) with adequate measuring systems have to be provided at the supplier's premises for independent evaluation and agreement.

The supplier, in cooperation with AUDI AG Logistics, has to prepare a packaging schedule two months before PVS, that complies with product-specific requirements. The supplier is responsible for providing parts that are clean and free of damage.

For tracing planning and implementation activities, the supplier has to apply the Qualification Programme for New Parts (QPN) (= Formel Q New Parts – integral) (*Attachment 1*) during the entire product and process development phase. Ratings and measures have to be submitted to Quality Assurance at AUDI AG in accordance with the rules specified in QPN for each priority key figure.

2.2. Quality planning series production

Continuous verification, evaluation and updating of the process flows, taking into account any problems encountered, has to be conducted during the development, start-up and production phases with the aim of continuous improvement on the part of the supplier. Experience gained in production with the product and process, has to be stored in a sensible form (e.g. in FMEA's) to permit it to be used in future projects.

3. Parts lifecycle, identification of parts, quality certificate for pre-production

An unequivocal and durable assignment of the combination of supplier, production facility, AUDI part number as well as production date has to be assured.

3.1. Parts lifecycle

A parts history (*Attachment 2*) has to be maintained with the commencement of first production for the pre-series phase. All part modifications and optimisations have to be listed therein. Repeat supplies with the same status shall not be entered. The part history has to be submitted together with the quality verification at every change in the pre-production.

3.2. Identification in the pre-series phase

An unequivocal parts identification is required on each part for each delivery as from the first prototype in order to ensure unequivocal assignment of each part supplied. Furthermore the outside of the delivered unit/package is to label with a explicit note: "KEINE SERIENTEILE" (no series parts).

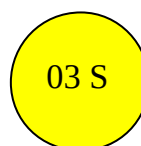
If the type of identification specified in the drawing is not yet possible during the pre-series phase, an adhesive label has to be used in this case. This label has to contain the name of the supplier, the number and description of the part, the date of production as well as the status of the drawing or agreement.

If the safety technical function has not been granted for safety-relevant features such as airbag module, dash panel, post trims, seats, steering wheels, pre-tensioners amongst others, then this "Non-function" or "Non-approval of prototype" has to be clearly and permanently labelled so that it can be retraced. This also applies for the "Non-function" of individual components or a sub-assy. Labelling has to be recognisable even when the component has been installed and has to be approved by QA Product Technology/Purchased parts at AUDI AG beforehand.

In addition, during the pre-production phase, a yellow circular adhesive label (see illustration below) indicating the status of agreements (consecutive numbering) and the tooling information (also applies to first or advance sampling) has to be affixed to the rear of all parts. Sheet metal parts, which are painted during vehicle assembly, has to be mechanically labelled (see LAH 893 012, AUDI AG manual for forming technology). In the case of small parts which are too small for affixing an adhesive label, the smallest possible packaging unit has to be marked. The hardware/software status (HW/SW status) has to be always stated for electronic components.

Example: Generation status (Gen.) 03 S in the yellow circular adhesive label

03 = 3rd modification (refers to drawing
or agreement status and parts lifecycle)



S = Production tooling

K = Small series tooling / Individual series tools (customer standard pre-production tooling)

H = Pilot tooling/Hand-made samples

Precedence of the generation states for example 01 H, 02 H, 01 K, 02 K, 03 K, 01S, 02S,...

Examples of the generation state:

- 01H = partial generation 01 of pilot tooling,
- 02H = partial generation 02 of pilot tooling,
- 03K = of partial generation 03 from limited-lot production tool or parts from unconcatenated tools/ customer standard pre-production tooling
- 01S = partial generation 01 from serial tool.

As a general rule, parts for PFF, PV and 0-series always have to be gauged. The measurement has to be recorded statistically and evaluated. The evaluations have to be submitted at regular intervals to the technical department responsible. The technical departments may, if need be, request individual documentary proof.

The labelling has to be removed when the preproduction phase ends (also first sampling of the parts from production/small series tooling has to be submitted at at least grade 3) and the parts are no longer supplied to the special warehouse.⁴

3.3. Quality certificate for the pre-production phase

For each supply of parts to the special warehouse, the supplier has to include the "Quality certificate for the pre-production phase" (*Attachment 3*).

In the case of electronic parts, the hardware / software status (HW/SW status) has to also be supplied always.

The quality of the parts delivered is assessed by the supplier as followed using the original drawings as a basis:

Dimensional check:	1 ☐	3 ☐	6 ☐
Material check*:	1 ☐	3 ☐	6 ☐
Functional check:	1 ☐	3 ☐	6 ☐

*When carrying out the material check on components for the interior furnishings and fittings, the status of the graining and colour specification has to also be taken into account for this evaluation.

Grade 1: No deviations from the technical documentation or only slight that, however, have no influence on function, look or assembly components

Grade 3: Parts deviating from the technical documentation, thus having a negative influence on processing, installation, functionality or visual characteristics. The features to be corrected must be specified.

Grade 6: Parts with grave deviations from the technical documentation, thus considerably hampering / affecting processing, installation, functionality or visual characteristics. The features to be corrected must be specified.

⁴ For the running series as well as for the customer support in the field it would be helpful if besides to the partial marking the partial generation state was a firm component of the marking.

All specifications indicated on the drawing, norms and technical supply conditions have to be fulfilled through the supplier. Post-treatment and deviations with important features must be agreed on with the responsible of Audi.

The supplier has to document all information on the agreement state with mention of the contact AUDI in the proof of conformance to specification „Qualitätsnachweis“ (for example members from the technical or the quality department).

The quality certification for the preproduction phase has to end when the preproduction phase ends (also first sampling of the parts from production/small series tooling has to be submitted at at least grade 3) and the parts are no longer supplied to the special warehouse.

Further demands and references to the procedure in the standard "Proof of conformance to specification for the Series before stage" / „Qualitätsnachweis für die Vorserienphase“ (see applicable documents).

3.4. Marking at changes in the series

The supplier has to sign the delivery with the change triangle (see attachment 8) to every Audi-Plant obliged in the case of notifiable changes. The marking is complete and filled at every delivery unit clearly visible and protected against harm or loss. With the BTV/T of AUDI the escape of the marking has to be agreed.

4. Master jig harmonisation

As early as two months before PVS, the current parts has to be submitted by supplier at his own volition, following agreements with AUDI AG Quality Assurance-Product Technology/Purchased parts, for assessment on the interior or exterior master jig or on the function cubes. All components which are installed at a visible point in the interior of the vehicle and the related attachment elements, has to be assessed on the interior master jig. Body or body attached parts has to be assessed on the master jig or function cube.

The supplier has to be solely responsible for optimising and harmonising his scope of supply in respect of function, fit, optics, haptics, odour characteristics, unobstructed operation, assured ease of assembly, ease of operation (forces, noises...) and stiffness. The supplier has to be responsible for coordinating the optimisation points with the sub-suppliers.

A competent contact has to be appointed who has to participate at regular intervals at the master jig discussions and has to be responsible for implementing the points agreed at the master jig.

Current samples, always in duplicate (colour and finish approved by Metrology / Master Jig department), has to be provided continuously for the master jig and cubing.

More than 2 parts may also be requested in certain circumstances.

4.1. Requirements for master jig harmonising

- Parts manufactured from Production tooling or Small series tooling / Individual series tools

- Parts with appropriate identification
- Parts with appropriate measurement report with:
 - Data on attachment and reference points
 - Data on connection and functional contours
 - Verification of dimensional tolerance by means of set/actual comparison
 - Information on the set-up

The deviations have to be shown in a projection either in coordinates according to mathematical symbol rules or in the direction of the normal (+ too much on part, - too little on part).

When illustrating mating contours, such as by using cross sections, it is necessary to scan the shape of the mating regions and to illustrate the set/actual comparison of the data in a 1:1 plot.

As a general rule the data for the measurement report has to be processed and illustrated (schematic diagrams). Columns of figures will not be accepted.

4.2. Conducting measurements and measurement procedure

The parts have to be measured once with original attachments and once free of tension. The reference point system (RPS) as specified in VW 01055 has to be observed. The position and number of measuring points has to always be approved before measurement by AUDI AG Quality Assurance Metrology/Master Jig department.

4.3. Data flowback

Following a passed approval on the part of the AUDI AG Quality Assurance Metrology / Master Jig department the harmonised data has to flow back without delay into the CAD data record of the design department responsible (supplier/customer).

4.4. Tooling/part design

Removal of parts from moulds after graining (in compliance with standards of Technical Development) has to be inspected in detail as a general rule and specifically after any change made on the part of the supplier. The parting lines in tooling has to be defined optically and agreed with Technical Development in the initial stages of tooling design as part of the SE work. Approval of the graining has to be given in writing by AUDI AG Quality Assurance Product Technology/Purchased parts after agreement by Quality Assurance Metrology/Master Jigs, Quality Assurance Material Technology and Technical Development.

5. Process capability prior to and during series production

The important quality characteristics defined for the product (of relevance in terms of function, production and quality) has to be identified in the drawing. This definition has to as a general rule be performed as part of the SE work, if necessary with the aid of FMEAs and QFD analyses in cooperation between AUDI and the supplier. The characteristics from the range of in-

portant characteristics which has to be defined has to be those for which process capability investigations are to be conducted.

The supplier has to define the operating conditions necessary for a zero fault process and provide documentary evidence of the capability of these conditions. These must subsequently be described in the process steering schedule.

The following limit values has to apply to the continuous verification and documentation of the process capability for the defined important characteristics:

- Provisional process capability $Ppk \geq 1.67$
- Long-term process capability $Cpk \geq 1.33$ at start of PVS, with continuous subsequent improvement (aim: $Cpk \geq 1.67$)

VDA Volume 4.1 as well as the AUDI Guideline (*Attachment 4*) has to be taken as a basis for determining machine and process capability parameters. Both short-term as well as long-term capabilities have to be determined and have to be documented no later than prior to pre-production (2TP).

AUDI AG Quality Assurance recommends for statistical analyses the programme qs-STAT 3.x from Q-DAS GmbH, Eisleber Str. 2, D-69469 Weinheim, Tel. +49 (0)6201/3941-0, e-mail: q-das@q-das.de

In this connection we refer VW 10130 and VW 10131 who explain a more practice-oriented evaluation of the machine ability investigation and process capability for perceptible features also to the applicable new VW-norms. The checking facility ability should occur according to the general guide of the European automotive industry "Ability proof of measuring systems", VW 10119 can alternative by agreement to be used. The demands are meant for systems determining. Requirements for software or software in the vehicle see: LAH 893.909.A

6. Internal process and product audits

Internal process audits are carried out within the scope of continuous improvement to establish that the processes and procedures used conform to the standards and requirements. Alongside these, internal product audits are carried out to check if the product fulfils the required quality requirements. The audits are carried out either at regularly scheduled intervals or randomly (event-related). The weakpoints revealed by the results of the audit have to be defined with effective remedial measures set out in an improvements programme with time schedule and implemented. (For further information refer to VDA volume 6.3 and VDA volume 6.5.)

7. First sampling

7.1. First sampling and approval requirements

First samples have to be manufactured with standard production resources under series production conditions and have to be submitted to AUDI AG Quality Assurance Product Technology/Purchased parts on schedule (deadline specified by Purchasing).

All parts have to be supplied from standard tooling for PVS and the components must obtain a grade of 3 or better in the sampling process. Incomplete documentation and non-compliance with the technical specifications has to lead to the first sample inspection report being rejected (Grade 6). It is necessary for the supplier to carry out the re-sampling of components that were originally given Grade 3 or 6 in the first sampling. For 0-series, an OK sample inspection with Grade 1 must be achieved.

At least 5 parts with related first sample inspection report has to be submitted for first sampling. The exact number of parts have to be agreed in advance with the respective clerk in AUDI AG Quality Assurance Product Technology/Purchased parts. Semi-finished products and single parts of a component or scope of supply must also be submitted to AUDI AG on request (this applies throughout the entire product lifecycle).

First samples have to be provided in the case of:

- New parts
- Any technical change
- New and/or modified production facilities, tooling / replacement tooling
- Modified processes (internally or at a sub-supplier)
- Production relocation (internally or at a sub-supplier, new sub-suppliers)
- Lengthy suspension of production (> 1 year)

The performance and the extent of renewed sampling has to be clarified at an early stage where doubt exists with the respective clerk in AUDI AG Quality Assurance Product Technology/Purchased parts.

The production conditions during the production of sampling parts have to be documented. These have to form the basis for consideration during 2TP.

Additional expenses that may arise in connection with incomplete sampling documents have to be invoiced to the supplier.

If the sampling check conducted by AUDI AG does not confirm the test results documented in the first sample inspection report conducted by the supplier, and a rating of 3 or 6 is therefore allocated as a result of the sampling, the inspection costs incurred has to be debited.

If a process inspection at the supplier plant should be necessary, AUDI can debited all the costs (for example travel costs).

No such costs have to be debited in the event that all deviations from the technical documentation are listed and identified by supplier.

BeOn: Within the framework of the optimization of the first sampling process with suppliers and combine-working in the VW Group we introduced the online-sampling system BeOn.

For all Audi projects the first sampling process is only still allowed with the online-sampling system BeOnline way, the utilization of the system BeOn is duty for all new supply component. You find nearer information concerning utilization and release on the B2B-platform.

7.2. Contents of first sampling documents

The complete first sampling documentation has to be submitted to AUDI AG Quality Assurance. These must contain:

- First sample inspection report cover sheet completed in full conforming to VDA, Volume 2, Attachment 2
 - Indication of production location with the relevant supplier number (with index)
 - Parts Lifecycle (see 3.1)
 - A statement of the generation status, and for electronic components, the HW-/SW status
 - The attachment to the VDA cover sheet "First Sampling" (*Attachment 5*) completed in full
 - A binding Declaration of Conformity to EC Guidelines (*Attachment 6*) completed in full
- Full details relating to material composition (material data) of all individual parts complying with IMDS in accordance with VW Standard 01155 as well as specification of the IMDS identification number on the first sample inspection report ([flyleaf under remarks](#)).
If the information should still be missing in the IMDS, a declaration of conformity (*attachment 6*) for the documentation of the compliance of the old vehicle law must be given in a temporary way. The IMDS-data must be handed later up to the end of the supply component as samples at AUDI. As long as any material data accepted by LIMITED COMPANY AUDI there are not in the IMDS, the first model is evaluated at best with a note 3, in this case a Nachbemusterung zwingend is necessarily. A later determination of the non-fulfilment of the declaration of conformity in a temporary way exhibited and material data not accepted by LIMITED COMPANY AUDI lead 6th to some supply component as samples with the note
- - EC Safety Data Sheet for Process Materials and Hazardous Substances (if applicable cf. VDA Volume 2, Attachment 2)
 - Measurement report completed in full together with drawing or outline sketch with identification of the measuring points - not figures in columns ! All non-compliance characteristics, such as dimensions not to tolerance or specifications not met in the functional test, have to be agreed in advance with the respective designer. Deviations from the technical documentation have to be identified by the supplier in the first sample inspection report (on the cover sheet, on a separate sheet or on the drawing by the supplier using the set/actual comparison (illustration of difference)). If Technical Development has issued an allowable deviation (AE), this must be enclosed.
 - Complete inspection reports about every single part inspection, functional test, surface test etc.
 - Process capability certificate for the defined important characteristics
 - The sub-supplier has to be identified for the first sampling of assemblies. Complete proof of the sub-suppliers' individual part sampling inspections have to be submitted and enclosed in the sampling documentation for AUDI AG. For sampling assemblies with set parts which are sampled by AUDI AG (e.g. textiles/leather), complete proof of the OK sample inspections of the set parts have to be enclosed with the sample inspection documentation for AUDI AG. The sub-supplier's complete sampling inspection documentation has to be made available to AUDI AG Quality Assurance, on request of AUDI AG.

Reference should be made to the VDA first sample inspection report cover sheet for further possible attachments.

The first sample inspection report has as a general rule be agreed in advance with the respective clerk in Quality Assurance Product Technology/Purchased parts AUDI AG.

First sampling is recommended for parts with critical deadlines in order to reveal any problems at an early stage. Supplier has to identify advance samples by a V on the first sample inspection report.

7.3. Addresses where first samples are to be sent

The first samples must be supplied on schedule to the responsible factory mentioned from the purchase with the order for the supply component as samples in the requested amount. The supply component as samples documents must be complete (s.a. ch. 7.1. concerning online-sampling with BeOn). Additional expenditure, that in connection with an incomplete delivery of the requested first samples, the supply component as samples documents or delivery at the wrong site / the wrong unloading time, the supplier is charged. The supplier has to inform fundamentally the duty of delays the customer in time.

Ingolstadt	Neckarsulm	Győr	Quattro GmbH
AUDI AG Ingolstadt Halle N52 I/GQ-462 D-85045 Ingolstadt	AUDI AG Neckarsulm Geb. B9-Rampe N/GQ-546 D-74148 Neckarsulm	direct: AUDI HUNGARIA MOTOR Kft. Abladestelle SL00 Halle G14 Kardán út 1. H-9027 Győr or AUDI-transport: AUDI AG Ingolstadt GVZ Halle F Weiterleitung AHM Hr. Pergel, G/GQ-3 Abladestelle SL00 D-85045 Ingolstadt	Quattro GmbH Neckarsulm Geb. B6 Wareneingang N/VQ-Q Herr Lunczer 74148 Neckarsulm
FAW-Volkswagen			
Mr. Li Shuang Quality Assurance QA-PP Dongfeng Street CN-13011 Changchun P.R. of China			

8. Pre-production (2-days' production)

Pre-production has to be carried out prior to the commencement of series production in order to provide verification that the parts comply with the qualitative and quantitative standards at the start of production and in series production. The "Qualification Programme for New Parts (QPN) (= Formel Q New Parts – integral)" (*Attachment 1*) must be successfully completed for this purpose.

For pre-production a quantity which is adequate for statistical analyses (P_{pk}/C_{pk}) has to be produced continuously at series production speeds and packed in the correct standard packing provided for this purpose. Evidence of the process capability for important specified characteristics has to be furnished no later than preproduction (2TP) (see 5). The exact date and the production batch size has to be agreed in advance with the customer. Pre-production has to be carried out at a sufficiently early stage to enable necessary process improvements to be made prior to the start of series production, i.e. as close as possible to 0-series production (*Formel Q-konkret* Section 3.2.5 between 0-series and start of production).

Pre-production has to be carried out by the supplier at his own volition and the result advised to Quality Assurance Product Technology/Purchased parts AUDI AG. When it comes to parts of major importance as defined by AUDI AG (examples being modules of complicated components, assemblies or components) the respective officer for Quality Assurance Product Technology/Purchased parts AUDI AG, as need be with the participation of Technical Development, Quality Assurance Laboratory, Purchasing and Production Planning, may carry out acceptance inspection. If major weakpoints are revealed in pre-production which necessitate a repetition, the costs of the repetition of the pre-production shall be borne by the supplier.

Requirements for acceptance of pre-production

- Operating resources installed in conformity with series production
- Build samples and first samples released for series production

Provided the objectives are attained, approval for series production has to be issued by the Technical Officer Quality Assurance Product Technology/Purchased parts AUDI AG.

9. Quality backup during series production

9.1. 0-km complaints

Faulty purchase parts have to be analysed without delay, remedial measures have to be defined for all causes of faults, scheduled, applied and verified in terms of their effectiveness. This includes all necessary measures at the sub-supplier's premises.

As a rule, the revision of the process FMEA at the sub-supplier's premises has to be ensured for every reclamation and submitted to AUDI AG Quality Assurance to avoid repetitive errors.

The processing of the complaint has to be performed using the generally available QTS information and processing system. The AUDI AG Quality Assurance Technical Officer has to be advised **without delay** (< 24h) in writing, e.g. by means of an 8-D Report, of the action taken. The 8-D report has to be concluded within the agreed timeframe. Should it not be possible to maintain this time schedule, an interim report has to be submitted after this time period. [If the supplier can not keep to the set dates of the complaint , he has to agree on the analysis time](#)

needed by him (Date) with the BTV/TV immediately. Should deadlines be missed during the processing of reclamations about purchased parts (8-D report) that have not been approved by AUDI AG Quality Assurance, the 8-D report will be concluded after 4 weeks at the supplier's expense, irrespective of the question of cause.

If 0-km faults occur in purchased parts due to supplier's negligence, any costs incurred (including ancillary costs such as moving the vehicle, handling parts, moving material etc.) shall be charged back to the supplier. The charge-back has to be based on the quality service costs of 0-km / field complaints (*Attachment 7*).

In consultation with AUDI AG Quality Assurance, quality reports in respect of the current ppm figures has to be provided at regular intervals.

9.2. Reworking and sorting action

In the event that defects in purchase parts are discovered within the Audi Organisation, the supplier has to assure on-the-spot technical support immediately or no later than within 24 h after the occurrence of the defects in order to be able to carry out any reworking or sorting action on-the-spot.

If repeated complaints occur in respect of purchase parts, AUDI AG Quality Assurance may request supplier to provide a Resident Engineer. The remaining warranty provisions shall not be affected by this.

9.3. Analysis of field complaints

In respect of faulty purchase parts reported from the field (customer complaints from authorised workshops) in accordance with the QTS System supplier has to ensure that the parts are analysed with the shortest possible delay, the necessary measures taken and the effectiveness of the measures verified. This also includes any measures which may be necessary at sub-suppliers'. *If the supplier can not keep to the set dates of the complaint , he has to agree on the analysis time needed by him (Date) with the BTV/TV immediately.*

As a rule, the revision of the process FMEA at the sub-supplier's premises has to be ensured for every reclamation and submitted to AUDI AG Quality Assurance to avoid repetitive errors. The processing of the complaints has to be performed using the generally available QTS quality data information and processing system. The AUDI AG Quality Assurance Technical Officer shall be advised **without delay** in writing, e.g. by means of an 8-D Report, of the action taken. The 8-D report has to be concluded within the agreed timeframe. Should it not be possible to maintain this time schedule, an interim report has to be submitted after this time period. Should deadlines be missed during the processing of reclamations about purchased parts (8-D report) that have not been approved by AUDI AG Quality Assurance, the 8-D report will be concluded after 4 weeks at the supplier's expense, irrespective of the question of cause.

If field complaints occur in purchased parts due to supplier's negligence, any costs incurred (including ancillary costs such as handling parts, moving material etc.) has to be charged back to the supplier. The charge-back has to be based on the quality service costs of 0-km / field complaints (*Attachment 7*).

Statistics have to be produced on the basis of field monitoring which shall be taken as a basis for deriving forecasts in respect of failure probabilities. An early warning system which permits countermeasures in respect of field complaints to be taken at an early stage, has to be installed at start of production.

Technical factors (TF) are determined by Quality Assurance based on parts analyses and assessments of shut-down measures between QA Purchased parts and supplier, these being recorded in the QTS System by means of inspection reports. The TFs reflect the supplier's responsibility. These TFs as well as the location of the mutually agreed range of parts are incorporated in the supplier agreement. The technical factor is entered in the debit note for a defined chassis number period for the commercial processing.

9.4. Continuous quality monitoring

Each supplier has to be familiar with how his products behave during their useful life. The supplier has to provide proof of the activities he has undertaken in order to obtain information regarding the quality of the products during their useful life.

To permit the supplier to conduct product monitoring, the QTS system has to be made available by Audi AG to the supplier as a source, over and above the existing quality discussions.

The supplier must guarantee series-accompanying the compliance of the specified demands (for example accuracy to size) of his components and have to document it. The proof is to be provided on request. The extent and frequency of the measurements and checking process have to be agreed with the BVT/TV.

9.5. Audits, Problem analyses and Technical Revision Supplier (TRL)

In the event that AUDI AG conducts an audit or a problem analysis at supplier's because the latter has demonstrably not complied with AUDI AG quality requirements (e.g. in the case of field or 0-km complaints), the costs incurred in such a case by AUDI AG may be invoiced to the supplier.

The same shall also apply when it is discovered that an improvement programme has not been or insufficiently implemented in the course of subsequent or repeat audits and the A-classification thus required (>92%) has not been achieved.

If during a TRL will be a need to define direct actions at supplier's or the result of the TRL will be "RED", the travelling costs and other costs in such a case by AUDI AG may be invoiced to the supplier (see Formula Qability).

9.6. Media

Appropriate media resources has to be maintained for rapid problem-oriented processing of complaints as well as for improving the exchange of communication and information (e-mail link with encryption facility based on open-PGP standard, connection to AUDI QTS System, Microsoft Office compatible EDP programmes; video conferencing systems based on international standard H.320, T.120 – may be requested by AUDI AG if necessary).

9.7. External service providers

All the facilities necessary for the analysis have to be available at the supplier's. In exceptional cases, external service providers may be called in to analyse problems (universities, institutes, external laboratories). External service have to be accredited for special services concerning ISO/IEC 17025:2000 or need a special certification by AUDI AG. For the checking process the supplier is responsible.

The names of external service providers whom it is planned to use in analysing problems, has to be given to AUDI AG at the start of the project.

Proof has to be provided that appropriate capacities are available. The supplier has to ensure that AUDI AG employees or their authorised representatives are provided with access to the business premises of the service providers selected during normal business hours to the extent that this is necessitated by the processing of the problem. Access has to be granted to them at any time if a hazard exists.

10. Attachments

Respect! Not all attachments are available in English language.

Attachment 1: Qualification Programm for New Parts (QPN) (= Formel Q New Parts – integral)

Attachment 2: Parts lifecycle

Attachment 3: Quality certification for pre-production phase

Attachment 4: Guideline for determining machine and process capability

Attachment 5: VDA cover sheet “First Sampling”

Attachment 6: Binding Declaration of Conformity to EC Guidelines

Attachment 7: Quality Service Costs 0-km / field complaints

Attachment 8: Change triangle for marking of deliveries „Änderungsdreieck für Kennzeichnung von Lieferungen“

Attachment 9: Regulation for the entry of data with complex parts and systems

Attachment 10: Demands on the manufacturing and checking-technique of electrical and electronic units „Anforderungen an die Fertigungs- und Prüftechnik von elektrischen und elektronischen Baugruppen“